

D: HPV-11

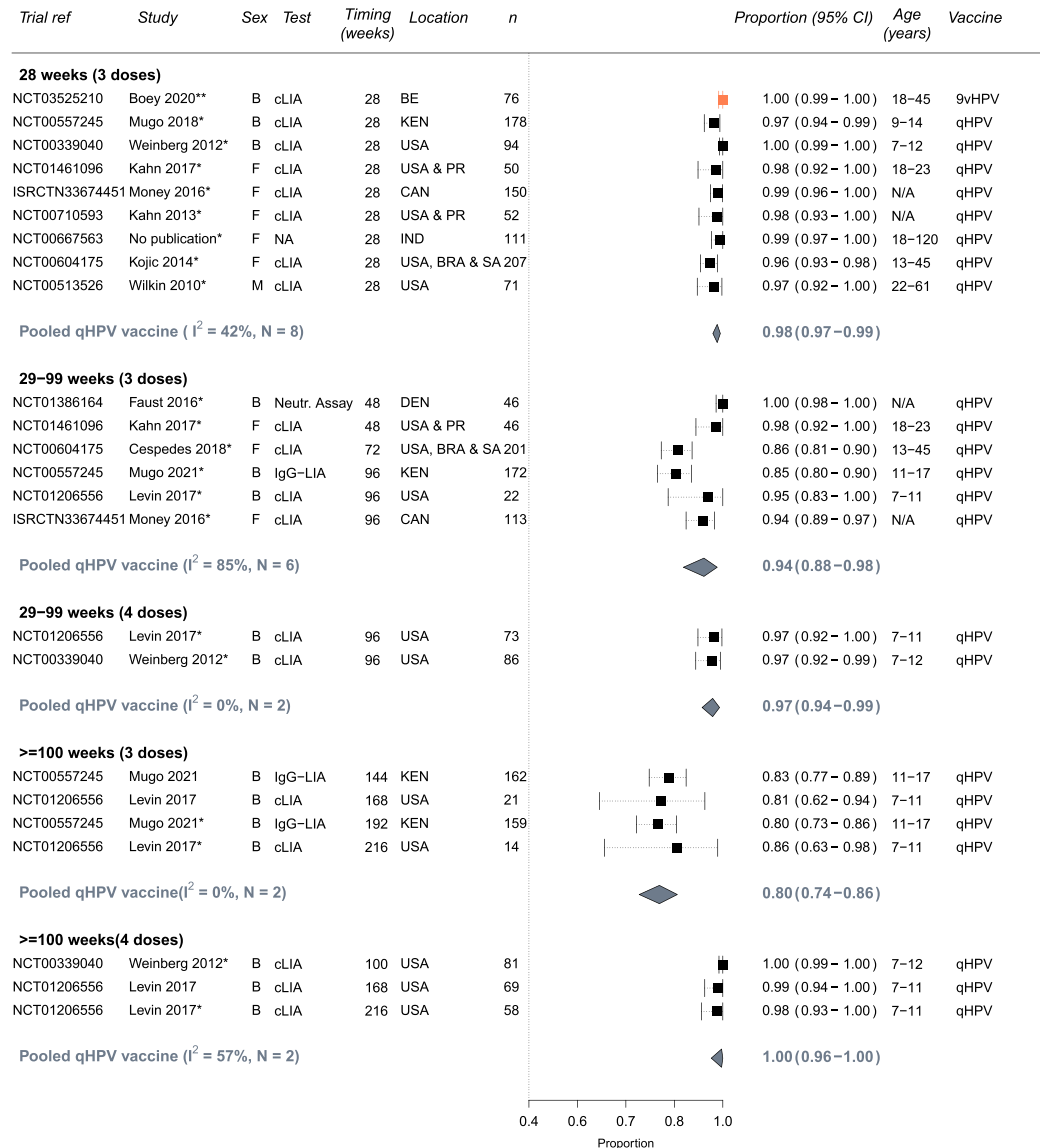


Figure 2. Continued

Table S3).^{48,89} A second trial reported a positive and significant correlation between high CD4% and antibody titers but did not provide stratified estimates⁵⁴ whereas a third trial found no difference in antibody titers by CD4% nadir⁵¹ (Supplement Figure S6A–D). Finally, a fourth trial reported higher seropositivity rates and antibody titers in PLHIV on ART than not on ART^{52,53} while another one reported higher antibody titers for all four HPV vaccine types among PLHIV virally suppressed compared to non-suppressed.⁵⁵

Biological endpoints

Results on various biological outcomes among PLHIV following vaccination were available from two RCTs

with placebo arms,^{56,57} one RCT comparing the bivalent and quadrivalent vaccine,³¹ two trials with historical controls^{46,58} and three single arm trials without control,^{32,49,59} Studies were conducted among women ($N_s=2$), men ($N_s=3$), or both ($N_s=3$), and among study participants HPV negative at baseline ($N_s=7$) or with unknown baseline HPV status ($N_s=3$). (Supplement Table S6; additional details in S7). No results were available for children. No results by HIV disease stage or ART status were available. Estimates of biological outcomes were generally deemed of low quality following our study quality assessment. Main study limitations included failure to specify the HPV vaccine type present in abnormal cytology samples, define baseline HPV